EduPath — AI-Powered Learning & Assessment Ecosystem

Adaptive Diagnostic Testing • 1-on-1 Live Tutoring • Zero-Cost Puter.js AI • Enterprise Dual-Engine DB

🎓 Learner Space

IRT dynamic placement tests, curriculum milestone tracks, interactive live classroom hub, and instant targeted remediations.

Dynamic Difficulty

👩‍🏫 Teacher Center

Scope-isolated workspaces (e.g. Math/Grade 8), Puter.js AI question generator, drag-drop test builder, and misconception heatmaps.

Pedagogical Guard

👨‍👩‍👧 Parent Portal

Real-time child progress monitoring, multi-tone weekly AI summaries (encouraging, analytical, firm), and card checkout receipts.

Family Oversight

🛡️ Admin & Security

Real-time heuristic anti-cheating telemetry (blur/pacing), content version diffing with 1-click rollback, and dual-mode database engine.

Full Auditability

https://edupath.app/platform-overview

• Active Deployment Ready

100% Free	< 1 ms	4 Roles
Client-side Puter.js AI with zero API monthly fees	Simulated JSON Cache + PostgreSQL Connection Pooling	Role-Based Access Control (RBAC) with instant demo switcher

Confidential & Proprietary • Google DeepMind Agentic Suite

LIFECYCLE ARCHITECTURE

The Unified 7-Stage EduPath Learning Loop

Continuous Diagnostic Pipeline: From Baseline Profiling to Automated Mastery Reporting

STAGE 01
Smart Onboard
Grade level & target subject registration.

STAGE 02
Diagnostic
IRT-scaled baseline test uncovers weak points.

STAGE 03
Tutor Match
Explore certified teachers by specialty & rate.

STAGE 04
Booking & Pay
Interactive date/slot & simulated card checkout.

STAGE 05
Live Tutoring
HD WebRTC stream, digital board & live notes.

STAGE 06
Retention Quiz
5-question exit checkpoint verifies retention.

STAGE 07
AI Digest
Multi-tone reports delivered to parents & gradebook.

1. Continuous Diagnostic Engine

Evaluations never end with a static letter grade. Every assessment recalculates student mastery curves and calibrates next-step question difficulty.

2. Human Educator Empowerment

Teachers arrive at 1-on-1 sessions with an auto-generated radar of the student's exact algebraic or conceptual missteps, avoiding 15 minutes of discovery triage.

3. Automated Remediation Loops

Exit quizzes trigger immediate custom micro-drills on missed concepts without requiring the educator to hand-author individual homework assignments.

Adaptive Diagnostic Testing Engine

Item Response Theory (IRT) Scaling • Real-Time Timer • Question Palette • Secure Lockdown

Grade 8 • Mathematics Placement Exam ⏱ 14:22 remaining

Question 3 of 10 — Difficulty: Medium (Weight 1.5)

Solve for x in the linear equation:

`3(x - 4) + 2 = 2x + 5`

A. x = 11

B. x = 15 ✓ (Selected)

C. x = 9

D. x = -5

Palette: 1 ✓ | 2 ✓ | **3 (Active)** | 4 | 5 | 6 | 7

Save & Next →

⚡ IRT Dynamic Calibration

If the student answers correctly, the next question escalates to weight 1.7+. If they stumble, the engine dials down to 1.2 to verify foundational concepts.

🛡 Anti-Guessing Fairness

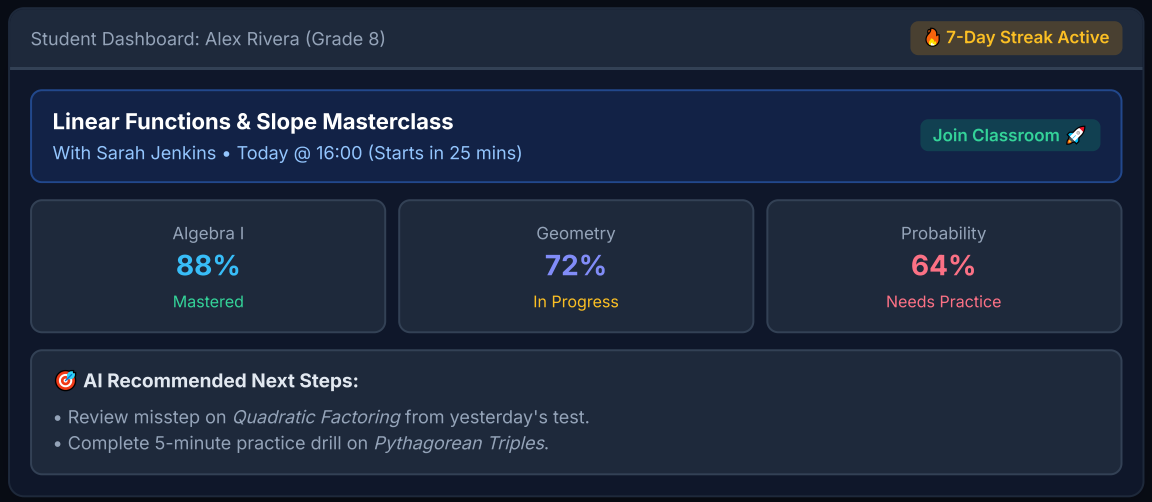
Students are given instant step-by-step mathematical explanations after submitting, turning diagnostic exams into rich learning opportunities.

📊 Immediate Skill Classification

Placement outputs pinpoint student percentile relative to national Grade 8 benchmarks, identifying remedial needs in linear algebra or geometry.

Student Dashboard & Milestone Mastery

Streaks • Upcoming Bookings • Personalized Curriculum Map • Quick Actions



🏆 Gamification & Retention

Daily streaks, Scholar levels, and achievement badges (First Ace, Speedster) drive continuous student engagement without arbitrary pressure.

📅 Instant Tutoring Teleportation

Learners access their scheduled virtual room with one click from their top banner without searching through external calendar links or email invites.

📈 Longitudinal Growth History

Progress timelines illustrate concept gains over time, reassuring learners that practice translates directly into measurable academic mastery.

Live Classroom Hub & Digital Notes Dock

Simulated HD WebRTC Stream • Interactive Whiteboard • Real-Time Agenda • Post-Lesson Checkpoint

• REC **Room #LSN-8042: Linear Functions with Sarah Jenkins**

⏱ 00:38:14 Elapsed • HD 1080p (18ms latency)



Sarah Jenkins, M.Sc.

WebRTC HD Video Stream Active

Alex (You)

```
Whiteboard: f(x) = mx + b  →  Slope m = (y2 - y1) / (x2 - x1)
```

📋 Session Agenda

✓ 1. Homework review (5m)

✓ 2. Slope formula derivation (15m)

• **3. Graphing exercises (Current)**

○ 4. 5-Question Exit Checkpoint

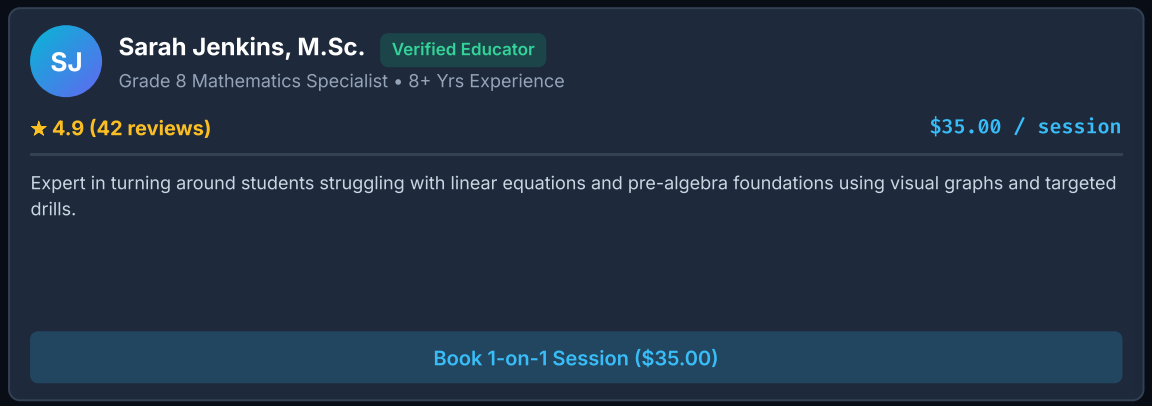
📝 Digital Notes Dock

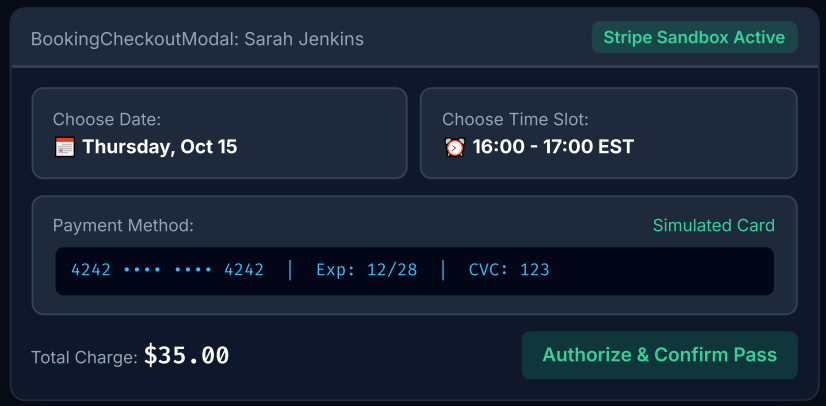
```
• Note: If m > 0, line rises.
• If m < 0, line falls.
• Parallel lines share identical m.
[Synced to Parent Portal]
```

Conclude & Launch Quiz

Teacher Selection & BookingCheckoutModal

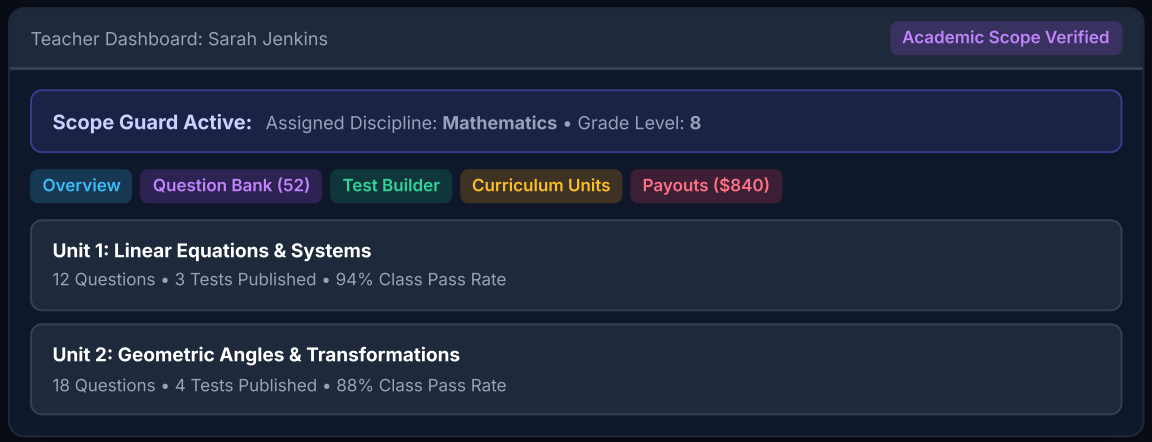
Verified Educator Search • Date & Slot Picker • Simulated Stripe Checkout • Session Pass Ledger





Teacher Command Center & Verification Guard

Strict Academic Scope Guard • Curriculum Planner • Lesson Materials • Direct Deposit Payouts



🔒 Zero Cross-Discipline Bleed

Teachers only create and grade content within their certified subject. A mathematics instructor cannot inadvertently edit history quizzes.

💰 Automatic Monthly Payout Ledger

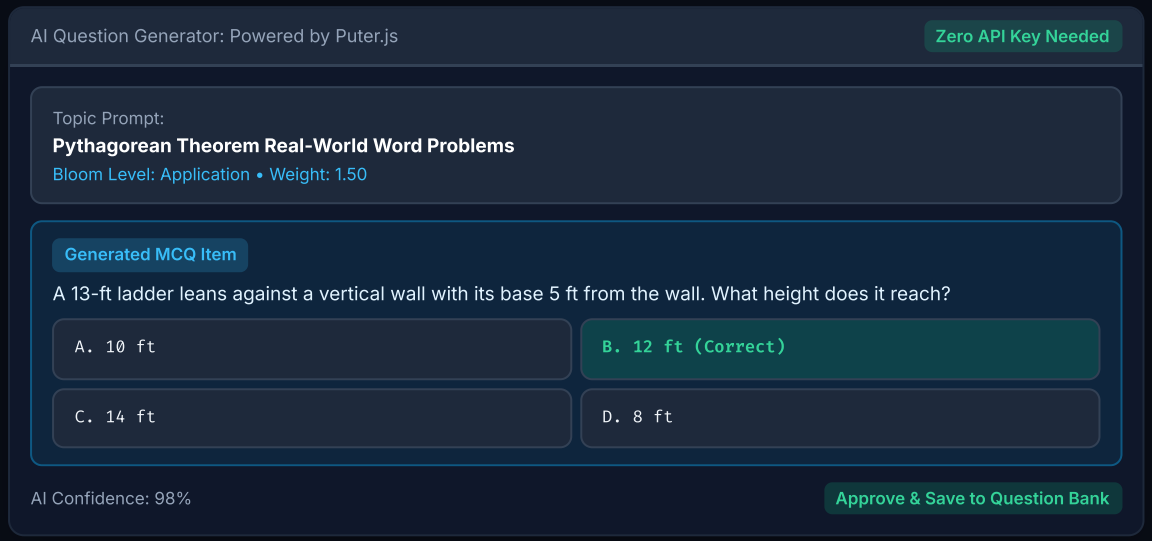
Completed live tutoring sessions credit directly to the teacher's earnings ledger ($35/session), with automatic monthly direct deposit payouts.

📂 Whiteboard Material Attachments

Upload PDF formula cheat-sheets and diagrams that automatically appear inside student digital notes docks upon classroom entry.

Question Bank & Puter AI Question Generator

Bloom's Taxonomy Weights • MCQ & True/False • Puter.js Zero-Cost AI • Bulk CSV/JSON Import



Rich Question Repository

Filter repository by difficulty (Easy, Medium, Hard), question type (MCQ, True/False), or search keywords in question bodies or explanations.

Bulk CSV & JSON Ingestion

Batch import 50+ questions from school district question files. Automatic schema validators prevent malformed choices or missing answer keys.

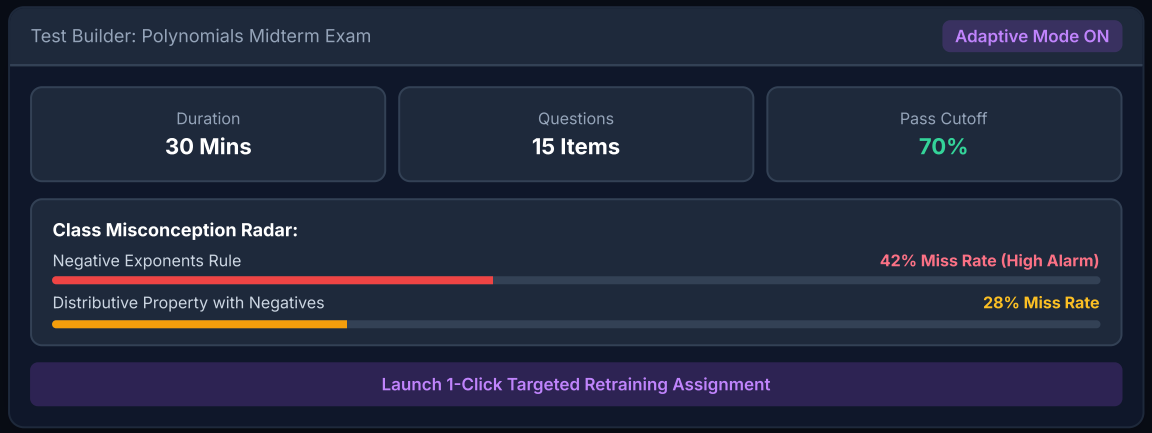
Mandatory Pedagogical Explanations

Every item includes step-by-step problem rationales ensuring students understand why their choice was correct or where their logic failed.

ASSESSMENT BUILDER

Adaptive Test Builder & Misconception Forensics

Timed Exams • Passing Thresholds • Item Misconception Radar • 1-Click Retraining Paths



🎯 Forensic Misconception Detection

Instead of reporting simple test averages, EduPath highlights specific conceptual blind spots that tripped up multiple students across the grade.

🔄 Automated Remediation Pipelines

One click automatically packages targeted 5-question micro-quizzes for students scoring below 70%, targeting only their individual missed items.

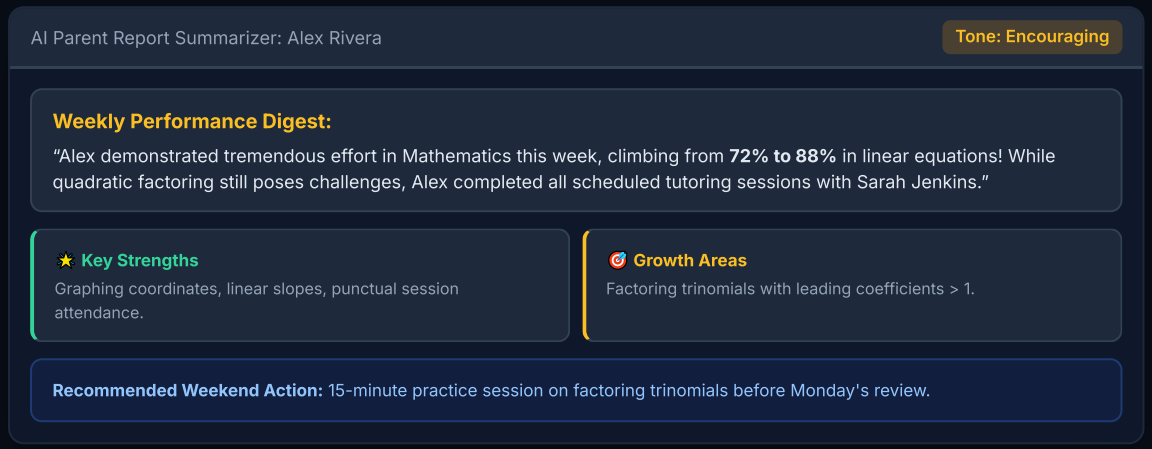
🔔 Zero-Delay Gradebook Notifications

Real-time submission alerts pop up immediately upon test completion, freeing educators from grading stacks of homework on weekends.

PARENT OVERSIGHT

Parent Portal & AI Weekly Report Summarizer

Linked Student Telemetry • Multi-Tone AI Summaries (Encouraging, Firm, Analytical) • Actionable Study Plans



🎙️ Customizable Tone Selector

Parents can toggle between *Encouraging* (positive reinforcement), *Analytical* (data and percentile metrics), or *Firm* (strict accountability reminders).

👨‍👩‍👧 Multi-Child Account Switching

Parents with multiple students enrolled can toggle between children seamlessly, reviewing individual teacher notes and scheduled appointments.

📄 1-Click Printable PDF Summary

Generate clean, professional 1-page reports to share with school counselors or private tutors during academic reviews.

AI Cheating Detection & Integrity Telemetry

Tab Blur Monitoring • Answer Pacing Anomalies (<3.5s) • Clipboard Traps • Triage Incident Workflow

Live Cheating Telemetry Monitor — 2 Incidents Flagged

Marcus Vance — Midterm Physics — High Severity

Trigger: Window Blur / Tab Switch (4 occurrences during exam)

Clear Incident | Require Retest

Elena Rostova — Grade 8 Algebra — Medium Severity

Trigger: Pacing Anomaly: 5 complex items answered in 7.2s

Clear Incident | Require Retest

```
Integrity Engine: Heuristic Rule-Based + Anomaly Scorer [Active]
```

👁 Non-Invasive Browser Telemetry

Monitors browser visibility state, fullscreen state, and clipboard paste listeners without invasive background webcam recording or invasive software installations.

⏱ Answer Pacing Forensics

Detects answer speeds faster than human comprehension thresholds (<3.5s per multi-step math problem) indicating script automation or lucky guessing.

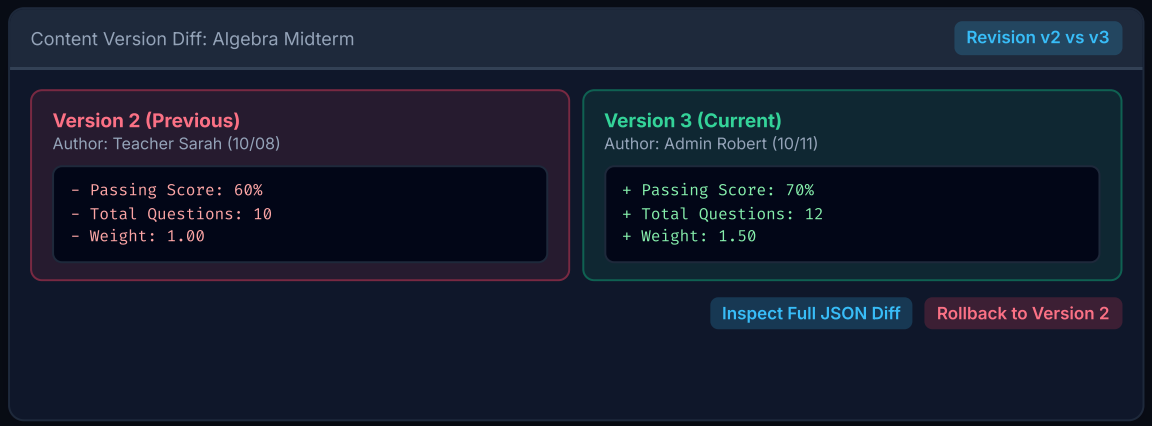
⚖ Administrative Due Process

Teachers and administrators can review timestamps, examine student response times, and decide whether to clear false positives or require a supervised retest.

CONTENT GOVERNANCE

Content Versioning & Full System Audit Trail

Reversible Test Edits • Side-by-Side Diff Viewing • 1-Click Rollback • Immutable Event Logging



⏪ Guaranteed Reversibility

If an educator makes accidental modifications to question keys or exam timings, admins can revert to any prior state instantaneously.

📜 Comprehensive Event Audit Trail

Tracks every login, privilege elevation, test submission, and question creation with full ISO timestamps and IP addresses.

🛡️ Multi-Tenancy Protection

Maintains curricular integrity across multiple schools and departments without risking database corruption.

INFRASTRUCTURE

System Diagnostics & Dual Database Architecture

Simulated JSON Engine ⟷ Enterprise PostgreSQL • Zero-DevOps Setup • Connection Pool Telemetry

System Telemetry & DB Mode Switcher — Dual Engine Ready

Active Mode	Query Latency	Total Records
Simulated JSON	< 1 ms	142 Seeded

PostgreSQL Pool Status: STANDBY (Ready on demand)

Client Data Hydrator: Local Sync + Event Bus

Trigger Safe Seeder

Switch to PostgreSQL Engine

⚡ Zero-Barrier Local Execution

Stakeholders can clone and run EduPath on any machine with just `npm run dev`. No database setup, Docker daemons, or credentials required.

🐘 Enterprise PostgreSQL Ready

Configure `DATABASE_URL` in environment variables, and the server switches to enterprise PostgreSQL connection pooling with transactional schema guarantees.

🔄 Automated Database Reseeding

Administrators can trigger mock reseeding in development mode at any time to return the platform to an initial state for clean evaluation demonstrations.

INTERACTIVE DEMO

Live Evaluation & Interactive Walkthrough Guide

Instant Demo Role Switcher • Direct Feature Navigation • Complete PDF Documentation Exports

1. Student Demo

Test taking, learning streaks, upcoming lessons.

`Route: /student-dashboard`

2. Teacher Demo

AI Question generator, test builder, gradebook.

`Route: /teacher-dashboard`

3. Parent Demo

Weekly AI digests, booking receipts, growth radar.

`Route: /parent-dashboard`

4. Admin Demo

Cheating telemetry, content diffs, DB health.

`Route: /admin-dashboard`

📑 Comprehensive Offline PDF Handbooks Available in Workspace Root:

EduPath_Student_Tutorial.pdf
Complete Student Handbook

EduPath_Teacher_Tutorial.pdf
Teacher Operations Manual

EduPath_System_Manual.pdf
4-Role System Manual

EduPath_System_Presentation.pdf
This 14-Slide Presentation